ECE 121B Four-page cheat sheet

Carriers, statistics and doping

Professor Emaminejad / Lectures 1–8

Reference values at 300 K

L1 p25; L3 p8–10

$$q = 1.602 \times 10^{-19}\,\mathrm{C}, \quad k_B = 8.617 \times 10^{-5}\,\mathrm{eV/K}$$

For V_T, use k_B = 1.381 × 10⁻²³ J/K with q in C.

$$V_T = \frac{k_B T}{q} = 25.85\,\mathrm{mV}$$

Thermal voltage in transport and junction formulas.

$$E_{g,\mathrm{Si}} \approx 1.12\,\mathrm{eV}, \quad \varepsilon_{\mathrm{Si}} = 11.7\varepsilon_0$$

Silicon gap and permittivity; ε0 = 8.85 × 10⁻¹⁴ F/cm.

$$N_C = 2.82 \times 10^{19}, \quad N_V = 1.83 \times 10^{19}\,\mathrm{cm^{-3}}$$

Effective band-state densities at 300 K.

$$n_i \approx 10^{10}\,\mathrm{cm^{-3}}$$

Rounded course value for silicon at 300 K.

1. Occupation and carrier statistics

L2 p15, 30–32; HW1 P1–2

$$f(E) = \frac{1}{1+\exp[(E-E_F)/(k_B T)]}$$

Probability that an electron state is occupied.

$$f(E_F) = \frac{1}{2}$$

At the Fermi level, regardless of temperature.

$$f(E_F + k_B T) = \frac{1}{1+e} \approx 0.269$$

One-kT occupation check; HW1 P2.

$$f(E) \approx e^{-(E-E_F)/(k_B T)}$$

Boltzmann tail when E − E_F ≥ 3k_BT.

$$1 - f(E) \approx e^{-(E_F-E)/(k_B T)}$$

Empty-state probability below E_F.

Density of states and thermal carriers

L2 p8–10, 26; L3 p6–12

$$g_C(E) \propto \sqrt{E - E_C}, \quad g_V(E) \propto \sqrt{E_V - E}$$

Allowed-state density near each band edge.

$$n = \int_{E_C}^{\infty} g_C f\, dE, \quad p = \int_{-\infty}^{E_V} g_V (1-f)\, dE$$

Count occupied conduction and empty valence states.

$$N_C = 2\left(\frac{2\pi m^*_{n,\mathrm{DOS}} k_B T}{h^2}\right)^{3/2}$$

Conduction-band effective density of states; use DOS mass.

$$N_V = 2\left(\frac{2\pi m^*_{p,\mathrm{DOS}} k_B T}{h^2}\right)^{3/2}$$

Valence-band effective density of states; use DOS mass.

> Use kg, J and s in the DOS equations; divide m⁻³ by 10⁶ to obtain cm⁻³.

$$N_{C,V}(T) = N_{C,V}(300\,\mathrm{K})\left(\frac{T}{300\,\mathrm{K}}\right)^{3/2}$$

Scale effective state densities with T; fixed DOS mass.

$$n = N_C e^{-(E_C-E_F)/(k_B T)}, \quad p = N_V e^{-(E_F-E_V)/(k_B T)}$$

Carrier densities in the nondegenerate limit.

$$n_i = \sqrt{N_C N_V}\, e^{-E_g/(2k_B T)}$$

Intrinsic concentration; HW1 temperature and gap changes.

HW1: compare intrinsic concentrations

HW1 P1; L2 p26

$$\frac{n_i(T_2)}{n_i(T_1)} = \left(\frac{T_2}{T_1}\right)^{3/2} \exp\left[-\frac{E_g}{2k_B}\left(\frac{1}{T_2} - \frac{1}{T_1}\right)\right]$$

Use for two temperatures when E_g is held fixed.

$$\frac{n_{i,2}}{n_{i,1}} = \exp\left(\frac{E_{g,1} - E_{g,2}}{2k_B T}\right)$$

Use for a bandgap change at fixed T.

2. Doping and charge neutrality

L2 p23–26; L3 p13–20; HW1 P3, P5

$$np = n_i^2$$

Mass action at thermal equilibrium only.

$$n + N_A^- = p + N_D^+$$

Charge neutrality; use ionized dopants.

$$\Delta N = N_D - N_A$$

Net donor concentration for compensated material.

$$n \approx N_D - N_A, \quad p \approx n_i^2/n$$

Majority and minority carriers for n-type; full ionization.

$$p \approx N_A - N_D, \quad n \approx n_i^2/p$$

Majority and minority carriers for p-type; full ionization.

$$n = \frac{\Delta N + \sqrt{\Delta N^2 + 4n_i^2}}{2}$$

Exact neutrality result near intrinsic or compensated conditions.

$$p = \frac{n_i^2}{n}$$

Recover the exact hole density at equilibrium.

> HW1 P5: net doping sets n or p; total ionized impurities N_D + N_A affect mobility.

Fermi-level location

L3 p11–20; HW2 P3

$$E_F - E_i = k_B T \ln(n/n_i)$$

Find equilibrium Fermi shift from electron density.

$$E_i - E_F = k_B T \ln(p/n_i)$$

Find equilibrium Fermi shift from hole density.

$$E_C - E_F = k_B T \ln(N_C/n)$$

Conduction-edge separation in Boltzmann regime.

$$E_F - E_V = k_B T \ln(N_V/p)$$

Valence-edge separation in Boltzmann regime.

$$E_i = \frac{E_C + E_V}{2} + \frac{k_B T}{2} \ln\frac{N_V}{N_C}$$

Intrinsic level when conduction and valence DOS differ.

$$E_V + 3k_B T \lesssim E_F \lesssim E_C - 3k_B T$$

Check nondegenerate approximation before using exponentials.

> Assumptions: thermal equilibrium, parabolic bands and full ionization unless noted. At high doping use Fermi-Dirac statistics.

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Motion, conductivity and continuity

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3. Drift, mobility and resistivity

L3 p26–41; L4 p4–22; HW1 P4–5

$$v_n = -\mu_n \mathcal{E}, \quad v_p = +\mu_p \mathcal{E}$$

Electron velocity opposes the field; hole velocity follows.

$$\mu = \frac{q\tau}{m^*}$$

Relate mobility to mean scattering time.

$$\ell = v_{\text{th}}\tau = v_{\text{th}}\frac{\mu m^*}{q}$$

Mean free path from thermal speed; HW1 P4.

$$\frac{1}{\mu} = \frac{1}{\mu_{\text{ph}}} + \frac{1}{\mu_{\text{imp}}}$$

Matthiessen rule; lower mobility dominates.

$$\mu_{\text{ph}} \propto T^{-3/2}, \quad \mu_{\text{imp}} \propto \frac{T^{3/2}}{N_D + N_A}$$

Estimate which scattering process changes with T.

$$\sigma = q(n\mu_n + p\mu_p), \quad \rho_{\text{res}} = 1/\sigma$$

Conductivity and resistivity from carrier densities.

$$R = \rho_{\text{res}} L/A$$

Convert material resistivity to device resistance.

> HW1 P5: convert dimensions to cm before using Ω·cm. Mobility follows total dopant concentration; resistivity curves assume one dopant type.

Drift and diffusion current

L3 p28–32; L4 p18–27

$$J_{n,\text{drift}} = qn\mu_n\mathcal{E}, \quad J_{p,\text{drift}} = qp\mu_p\mathcal{E}$$

Conventional current from field-driven carriers.

$$J_{n,\text{diff}} = qD_n\frac{dn}{dx}$$

Electron diffusion current from density gradient.

$$J_{p,\text{diff}} = -qD_p\frac{dp}{dx}$$

Hole diffusion current from density gradient.

$$J_n = qn\mu_n\mathcal{E} + qD_n\frac{dn}{dx}$$

Total electron current: drift plus diffusion.

$$J_p = qp\mu_p\mathcal{E} - qD_p\frac{dp}{dx}$$

Total hole current: drift plus diffusion.

$$\frac{D_n}{\mu_n} = \frac{D_p}{\mu_p} = V_T$$

Einstein relation for nondegenerate carriers.

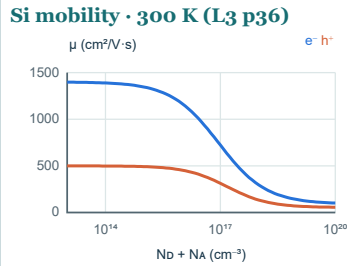
$$\mathcal{E} = -\frac{d\psi}{dx} = \frac{1}{q}\frac{dE_C}{dx}$$

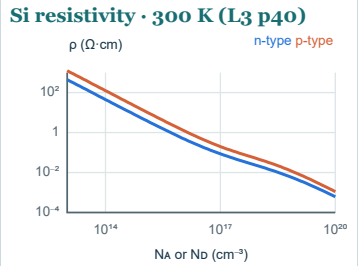
Electric field from potential or band-edge slope.

$$\mathcal{E} = -\frac{V_T}{n}\frac{dn}{dx}$$

Equilibrium field that cancels electron diffusion.

> For the band-edge slope, E_C must be in joules when q is in coulombs.





4. Continuity and generation

L4 p29–32; HW2 P1–2

$$\frac{\partial n}{\partial t} = \frac{1}{q}\frac{\partial J_n}{\partial x} + G - R$$

Electron balance in one dimension.

$$\frac{\partial p}{\partial t} = -\frac{1}{q}\frac{\partial J_p}{\partial x} + G - R$$

Hole balance in one dimension.

$$\frac{\partial n}{\partial t} = \mu_n\frac{\partial(n\mathcal{E})}{\partial x} + D_n\frac{\partial^2 n}{\partial x^2} + G - R$$

Expanded electron form: drift, diffusion, generation and loss.

$$\frac{\partial p}{\partial t} = -\mu_p\frac{\partial(p\mathcal{E})}{\partial x} + D_p\frac{\partial^2 p}{\partial x^2} + G - R$$

Expanded hole form for constant mobility and diffusivity.

$$\Delta n = \Delta p$$

Pair generation changes both carrier populations equally.

$$\Delta n = G_L\tau_n$$

Uniform steady low-injection photogeneration; HW2 P1.

$$n = n_0 + \Delta n, \quad p = p_0 + \Delta p$$

Total density during illumination.

$$\frac{\partial \Delta p}{\partial t} = D_p\frac{\partial^2 \Delta p}{\partial x^2} - \frac{\Delta p}{\tau_p} + G_L$$

Field-free low-injection excess-hole generation equation.

$$\frac{\partial \Delta n}{\partial t} = D_n\frac{\partial^2 \Delta n}{\partial x^2} - \frac{\Delta n}{\tau_n} + G_L$$

Electron counterpart for a p-type neutral region.

> Expanded forms assume constant μ and D. Low injection: excess carriers ≪ equilibrium majority density; in p-type Si compare Δn with p₀.

Minority diffusion and illuminated bar

L4 p29–32; L5 p5–17; HW2 P2

$$D_p\frac{d^2\Delta p}{dx^2} - \frac{\Delta p}{\tau_p} = 0$$

Steady, field-free, unilluminated interior.

$$L_p = \sqrt{D_p\tau_p}, \quad L_n = \sqrt{D_n\tau_n}$$

Distance excess minority carriers survive.

$$\Delta p(x) = Ae^{-x/L_p} + Be^{x/L_p}$$

General one-dimensional diffusion solution.

$$\Delta p(0) = \Delta p(L) = \gamma N_D$$

Boundary condition for equal end illumination.

$$\Delta p(x) = \gamma N_D\frac{\cosh[(x - L/2)/L_p]}{\cosh[L/(2L_p)]}$$

Symmetric bar profile satisfying both ends.

$$J_p(x) = -qD_p\frac{d\Delta p}{dx}$$

Hole current at an edge; differentiate the profile.

$$\Delta p(x) = \Delta p(0)e^{-x/L_p}$$

Single-edge decay into a long neutral region.

$$\Delta p(t) = \Delta p(0)e^{-t/\tau_p}$$

Time decay after generation stops; no spatial current.

$$J_p(0) = \frac{qD_p\gamma N_D}{L_p}\tanh\frac{L}{2L_p}$$

Hole diffusion current entering the equally lit bar.

> In HW2 P2, γ is stated as 10^{-4}. Use the stated boundary value when checking the worked solution.

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Quasi-Fermi levels and PN electrostatics

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5. Quasi-Fermi levels

L5 p21–23; HW2 P3

$$n = n_i e^{(F_n - E_i)/(k_B T)}$$

Electron density out of equilibrium.

$$p = n_i e^{(E_i - F_p)/(k_B T)}$$

Hole density out of equilibrium.

$$F_n = E_i + k_B T \ln(n/n_i)$$

Locate electron quasi-Fermi level.

$$F_p = E_i - k_B T \ln(p/n_i)$$

Locate hole quasi-Fermi level.

$$np = n_i^2 e^{(F_n - F_p)/(k_B T)}$$

Connect quasi-Fermi splitting to pair density.

> At equilibrium F_n = F_p = E_F. HW2 P3: low injection barely moves the majority quasi-Fermi level; high injection moves both.

Electrostatic potential and Poisson equation

L5 p24–25; L6 p17–20

$$\rho_{\text{ch}} = q(p - n + N_D^+ - N_A^-)$$

Local space charge, including ionized dopants.

$$\frac{d\mathcal{E}}{dx} = \frac{\rho_{\text{ch}}}{\varepsilon_s}$$

Field slope from charge density.

$$\mathcal{E} = -\frac{d\psi}{dx}, \quad \frac{d^2\psi}{dx^2} = -\frac{\rho_{\text{ch}}}{\varepsilon_s}$$

Potential slope and curvature.

$$\psi(x_2) - \psi(x_1) = V_T \ln\frac{n(x_2)}{n(x_1)}$$

Equilibrium potential difference from electron densities.

6. Abrupt PN junction: voltage and width

L6 p17–29

$$V_{bi} = V_T \ln\left(\frac{N_A N_D}{n_i^2}\right)$$

Built-in potential at equilibrium.

$$U = V_{bi} - V_A > 0$$

Voltage across depletion; forward PN bias lowers U.

$$W = \sqrt{\frac{2\varepsilon_s U}{q}\left(\frac{1}{N_A} + \frac{1}{N_D}\right)}$$

Total depletion width under applied bias.

$$W \approx \sqrt{\frac{2\varepsilon_s U}{qN_{\text{light}}}}$$

One-sided junction approximation.

$$W \propto \sqrt{V_{bi} - V_A}$$

Forward bias narrows; reverse bias widens depletion.

> Depletion approximation: fixed dopant charge dominates, mobile carriers are neglected inside W.

Units and polarity reminders

HW1 P4–5; HW2 P4

$$1\,\mu\text{m} = 10^{-4}\,\text{cm}, \quad 1\,\text{nm} = 10^{-7}\,\text{cm}$$

Convert geometry before R, W or diffusion calculations.

$$1\,\text{eV}/q \equiv 1\,\text{V}$$

Numerically convert energy barriers to voltages.

Abrupt PN junction: field and charge

L6 p21–29

$$N_A x_p = N_D x_n, \quad W = x_p + x_n$$

Charge neutrality and width split.

$$x_p = \frac{WN_D}{N_A + N_D}, \quad x_n = \frac{WN_A}{N_A + N_D}$$

Find depletion distance on each side.

$$|\mathcal{E}_{\max}| = \frac{qN_A x_p}{\varepsilon_s} = \frac{qN_D x_n}{\varepsilon_s} = \frac{2U}{W}$$

Peak triangular field; voltage is field area.

$$|Q| = qAN_A x_p = qAN_D x_n$$

Magnitude of charge on either side.

$$\mathcal{E}_p(x) = -\frac{qN_A}{\varepsilon_s}(x + x_p)$$

Field on the depleted p side: −x_p ≤ x ≤ 0.

$$\mathcal{E}_n(x) = \frac{qN_D}{\varepsilon_s}(x - x_n)$$

Field on the depleted n side: 0 ≤ x ≤ x_n.

$$U = -\int_{-x_p}^{x_n} \mathcal{E}(x)\,dx = \frac{|\mathcal{E}_{\max}|W}{2}$$

Voltage equals the triangular field area.

7. Depletion capacitance and C–V

L6 p9–11

$$C = \frac{\varepsilon_s A}{W}$$

Small-signal depletion capacitance.

$$i_{\text{ac}} = C\frac{dv_{\text{ac}}}{dt}$$

AC displacement current through depletion.

$$\frac{1}{C^2} = \frac{2(V_{bi} - V_A)}{q\varepsilon_s A^2 N}$$

Linear C^{-2} plot for a one-sided junction.

$$m = \frac{d(1/C^2)}{dV_A} = -\frac{2}{q\varepsilon_s A^2 N}$$

Slope yields the lightly doped density.

$$N = -\frac{2}{q\varepsilon_s A^2 m}$$

Extract doping from measured C–V slope.

> These C–V signs use the n-type forward-bias convention. The voltage-axis intercept is V_bi.

$$qV_{bi} = \Phi_{Bn} - (E_C - E_F)_{\text{FB}}$$
$$= \Phi_{Bn} - k_B T \ln\left(\frac{N_C}{N_D}\right)$$

For an n-type Schottky contact, use V_bi from the C–V intercept and N_D from its slope to find Φ_Bn; energies in eV.

Check before substitution

Lectures 5–6

> Use ε in F/cm, area in cm² and dopant density in cm^{-3}. Reverse bias increases W and lowers C. Mark p and n sides before assigning x_p and x_n.

Continuity: useful limiting cases

L4 p29–32; HW2 P1–2

> **Zero field / no drift:** $\mathcal{E} = 0 \Rightarrow J_n = qD_n n',\ J_p = -qD_p p'$. Diffusion can still flow.
>
> **Thermal equilibrium:** $\partial_t n = \partial_t p = 0,\ G = R,\ J_n = J_p = 0$. Drift and diffusion may cancel.
>
> **Uniform, steady light:** $0 = G_L - \Delta n/\tau_n \Rightarrow \Delta n = G_L \tau_n$ for minority electrons in p-type Si.

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Contacts, diode current and breakdown

Professor Emaminejad / Lectures 1–8

8. Metal–semiconductor contacts

L5 p27–39; L6 p4–8; HW2 P4

$$\Phi_{Bn} = \Phi_M - \chi, \quad \Phi_{Bp} = E_g - \Phi_{Bn}$$

Ideal barrier energies; Φ_M is metal work function, χ electron affinity.

$$\Phi_{Bn} = qV_{bi,n} + (E_C - E_F)$$

Barrier = band bending + bulk separation; every term is an energy (eV).

$$E_C - E_F = k_B T \ln(N_C/N_D)$$

Bulk n-type band-edge separation.

$$E_F - E_V = k_B T \ln(N_V/N_A)$$

Bulk p-type band-edge separation.

$$V_{bi,n} = \frac{\Phi_{Bn} - (E_C - E_F)}{q}$$

n-type built-in voltage from barrier and bulk Fermi level.

$$V_{bi,p} = \frac{\Phi_{Bp} - (E_F - E_V)}{q}$$

p-type built-in voltage from barrier and bulk Fermi level.

$$W_n = \sqrt{\frac{2\varepsilon_s(V_{bi,n} - V_A)}{qN_D}}$$

n-type contact: reverse bias V_A < 0 widens W.

$$W_p = \sqrt{\frac{2\varepsilon_s(V_{bi,p} + V_A)}{qN_A}}$$

p-type polarity used in L6: V_A > 0 is reverse bias.

> HW2 P4 realistic NiSi/Si values: electron barrier Φ_Bn = 0.65 eV; hole barrier Φ_Bp = 0.47 eV. Use these instead of ideal work-function estimates.

> At fixed metal and semiconductor, applied bias changes band bending and W, not the interface barrier height.

$$\frac{W_n(V_A)}{W_n(0)} = \sqrt{1 - \frac{V_A}{V_{bi,n}}}$$

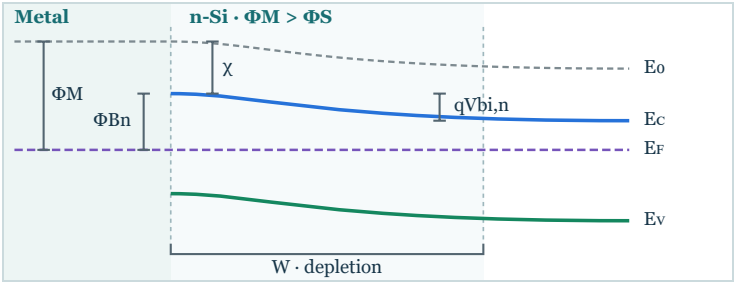
Compare n-type contact widths across biases; HW2 P4.

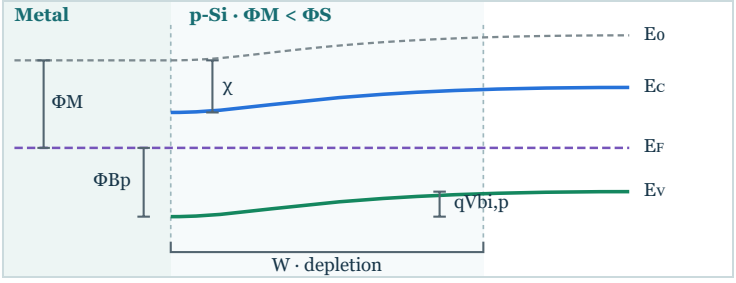
$$|Q_s| = qN_D A W_n$$

Depletion charge magnitude in an n-type contact.

Schottky barriers at equilibrium

L5 p29, 38; ideal rectifying contacts; schematic, not to scale





9. Ideal PN diode

L7 p6–20

$$n_{p0} = n_i^2/N_A, \quad p_{n0} = n_i^2/N_D$$

Minority equilibrium densities at each neutral side.

$$n_p(-x_p) = n_{p0}e^{V_A/V_T}, \quad p_n(x_n) = p_{n0}e^{V_A/V_T}$$

Minority boundary densities under applied bias.

$$J = J_0\left(e^{V_A/V_T} - 1\right), \quad I = AJ$$

Ideal current density and terminal current.

$$J(V_A \ll -V_T) \approx -J_0$$

Reverse saturation before breakdown.

$$J_0 = qn_i^2\left(\frac{D_n}{L_n N_A} + \frac{D_p}{L_p N_D}\right)$$

Reverse saturation density for long neutral regions.

$$\Delta p_n(x_n + x) = p_{n0}(e^{V_A/V_T} - 1)e^{-x/L_p}$$

Injected hole profile in the neutral n-side.

$$\Delta V_{\text{decade}} = V_T \ln 10 \approx 59.5\,\text{mV}$$

Ideal 300 K semilog forward-current slope.

$$r_d \approx \frac{V_T}{I}$$

Small-signal resistance at strong ideal forward bias.

> Ideal law assumes low injection, steady state, long neutral regions and negligible depletion-region recombination.

10. Nonideal current and breakdown

L8 p4–16

$$I_{\text{R–G}} \approx -\frac{qAn_iW}{2\tau_0}$$

Reverse generation current in depletion.

$$\tau_0 = \frac{\tau_p n_1 + \tau_n p_1}{2n_i}$$

Effective trap lifetime used in reverse generation.

$$R_{\text{net}} = \frac{np - n_i^2}{\tau_p(n + n_1) + \tau_n(p + p_1)}$$

Trap-assisted recombination; sign shows net R or G.

$$n_1 = n_i e^{(E_T - E_i)/(k_B T)}, \quad p_1 = n_i e^{(E_i - E_T)/(k_B T)}$$

Trap reference densities from trap energy E_T.

$$I_{\text{F,R–G}} \propto n_i W e^{V_A/(2V_T)}$$

Forward depletion recombination; ideality near two.

$$V_A = V_J + IR_S$$

Subtract series drop to find junction voltage.

$$V_{\text{BR}} \approx \frac{\varepsilon_s \mathcal{E}_{\text{CR}}^2}{2qN_{\text{light}}} - V_{bi}$$

One-sided critical-field estimate; not a current law.

> Avalanche: impact ionization, usually wider depletion region. Zener: tunneling through very narrow, heavily doped depletion region.

Last checks

Course conventions

> State assumptions, use consistent cm–F–V units, keep electrons and holes distinct, and check the sign of drift versus diffusion current.